

Quiz — Moment of a Force (Torque)

Time: 15 min
Total: 5 marks

Physics · Grade 11 · Multiple choice · 5 questions

Name: Class: Date:

Data: $\sin 30^\circ = 0.5$ | $\sin 60^\circ = 0.866$ | anticlockwise moment (+), clockwise moment (-).

Instructions: choose the **one** correct answer for each question and write its letter in the box.

Q1 A force of **40 N** is applied **perpendicular** to the handle of a spanner at a distance of **25 cm** from the centre of the nut (Fig. 1). The magnitude of the moment of the force about the nut is:

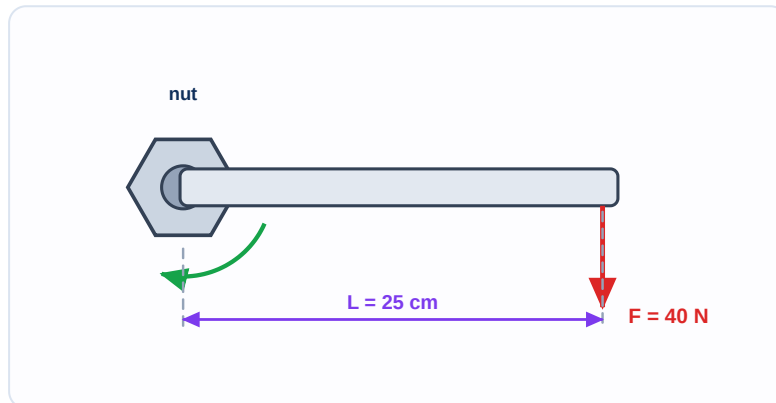


Fig. 1

A) 1000 N·m

B) 10 N·m

C) 160 N·m

D) 1.6 N·m

Q2 A force of **60 N** acts at the end A of a rod OA of length **1.5 m**, making an angle of **30°** with the rod (Fig. 2). The magnitude of its moment about O is:

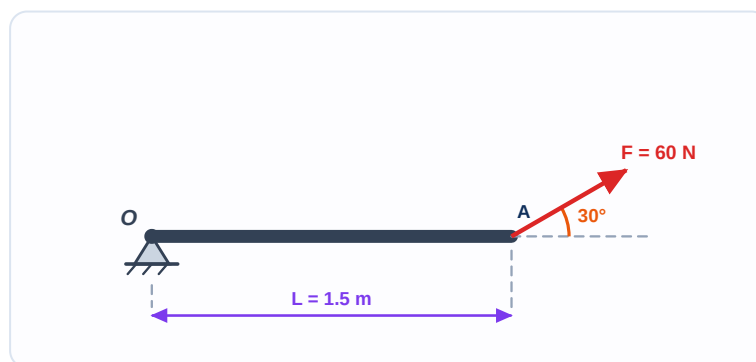


Fig. 2

A) 90 N·m

B) 45 N·m

C) 77.9 N·m

D) 30 N·m

Q3 A door is 0.8 m wide. Which of the following forces produces the **largest** moment about the hinge?

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A) 50 N perpendicular to the door, at its outer edge (0.8 m from the hinge)

B) 80 N perpendicular to the door, at its middle (0.4 m from the hinge)

C) 60 N at 30° to the door, at its outer edge

D) 100 N along the plane of the door, at its outer edge

Q4 Three forces act on the light bar shown in Fig. 3, which can turn about O. The **resultant moment** about O is:

★ challenging

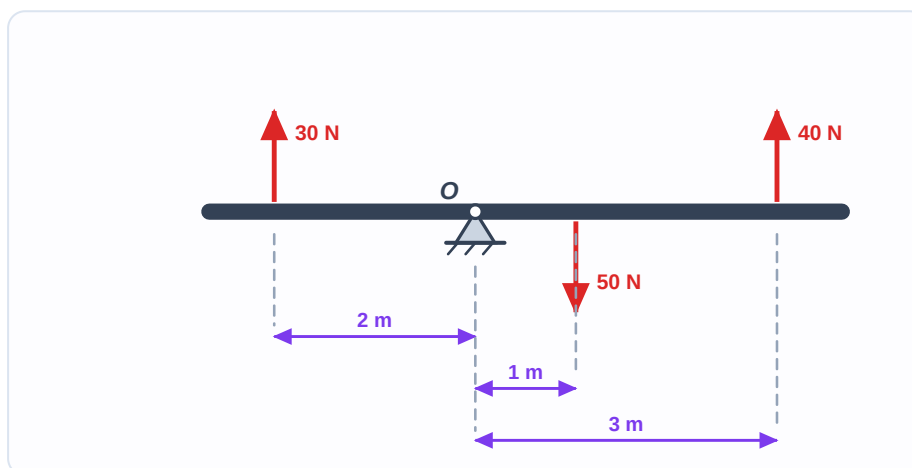
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Fig. 3

A) + 10 N·m (anticlockwise)

B) - 10 N·m (clockwise)

C) + 230 N·m (anticlockwise)

D) - 110 N·m (clockwise)

Q5 A force F acting at the end A of a rod OA of length 2 m, at an angle θ to the rod, produces a moment of **40 N·m** about O (Fig. 4). When the **same force** acts **perpendicular** to the rod at its **midpoint**, it produces a moment of **50 N·m**. The angle θ is approximately:

★ challenging

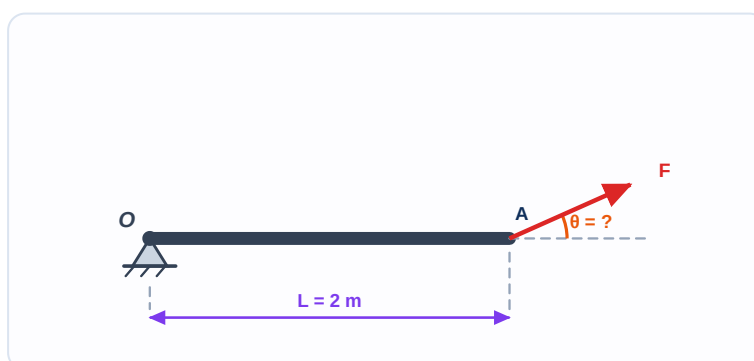
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Fig. 4

A) 23.6°

B) 53°

C) 30°

D) 66.4°